

EduScape AI

A Governed CBE AI Powered Tutoring and Learning Companion

Project period: Six months
Funding request: UGX 21,000,000

1. Problem Statement

As Uganda transitions to Competence-Based Education (CBE), universities must connect course materials, intended outcomes, assessment and timely student support.^[2] Yet a 2025 MUST study involving 237 respondents identified weaknesses in anytime access (2.88/5), outcome-aligned content development (2.77), timely availability (2.74) and sufficiency of information for tasks (2.52).^[1] This shows that making materials available online does not automatically provide effective, competency-focused support.

Ideally, every student should be able to work within their specific course offering, ask questions about lecturer-approved materials and receive immediate guidance connected to the relevant competency and learning outcome, with the supporting sources clearly shown. Lecturers should remain in control of course content, assessment and decisions about competency attainment.

MUST already has a functional Moodle-based VLE for materials, assignments, assessments, forums and interactive resources. However, students cannot easily ask questions about those materials and receive immediate answers supported by approved course content. Lecturers repeatedly address similar course-specific questions, while public AI tools do not reliably know the student's authorised materials, course context or intended outcomes. Misunderstandings can therefore persist, lecturers carry avoidable workload and students risk receiving unsupported or inappropriate answers.

The gap is not another LMS. What is missing is an intelligence layer that works with existing LMS platforms, connects lecturer-defined competencies and outcomes to approved resources, retrieves the right evidence, shows its sources and withholds an answer when evidence is insufficient.

EduScape AI will address this gap by complementing MUST's VLE while remaining adaptable to Moodle, Canvas, Open edX and other LMS platforms. Over six months, the project proposes to prepare and evaluate EduScape in two course offerings involving six lecturers and 100 students, subject to institutional and ethics approvals. Targets include at least 80% expert-rated answer correctness, 90% citation validity, 100% passage of cross-offering access-denial tests, median usability of at least 70 on the System Usability Scale, at least 80% completion of defined user tasks, median response time of no more than 12 seconds and a 30% reduction in document-to-roadmap preparation time.

2. Intervention / Innovation

2.1 Current Status Quo and Competitor Landscape

Existing alternatives solve parts of the problem. Moodle's core AI framework supports different AI providers, local models and controlled use within Moodle, but it does not itself provide EduScape's CBE roadmap and evidence cycle.^[3] LMS add-ons such as EduGears provide course-grounded tutoring, course creation and AI-assisted grading across several LMS platforms.^[4] Competency-oriented systems such as Artemis connect competencies, learner progress and recommended resources.^[5] Research tutors demonstrate citations and evidence-aware responses, while public AI assistants provide broad conversational support.

These alternatives are useful, but the reviewed descriptions do not show the complete combination of a lecturer-defined CBE model, an editable course-offering roadmap, roadmap-tagged knowledge, roadmap-aware student assistance, offering-level governance and continuing learning evidence.

EduScape combines these functions around one lecturer-controlled academic structure.

2.2 Detailed Description of the Innovation

EduScape AI is an LMS-agnostic intelligence and learning layer built around the lecturer-defined CBE framework. It has three connected layers:

1. **Institutional LMS layer.** The LMS remains the official source for identities, courses, enrolments, resources, submissions, grades and other academic records.
2. **Controlled integration layer.** EduScape establishes the user's identity, role, course offering and permissions, then maps authorised LMS information into a common format. Moodle is the first proposed integration. MUST ICT has verbally indicated willingness to provide an isolated test environment, subject to formal agreement.
3. **EduScape intelligence layer.** The lecturer digitises competencies, learning outcomes and assessment expectations. EduScape uses them to create an editable roadmap for the specific course offering. The lecturer reviews and controls the roadmap, which organises learning resources, activities, assessment and progress.

When approved notes are uploaded or imported, EduScape converts them into searchable course knowledge using Hierarchical Adaptive Retrieval-Augmented Generation (HA-RAG). Without exposing its detailed implementation, HA-RAG preserves both the precise passage needed and the wider section that gives it meaning. Lecturers can link materials to roadmap points, competencies and outcomes.

When a student asks a question, EduScape verifies access to that offering and considers both approved course evidence and the relevant roadmap context. It then provides a source-anchored response connected to the correct stage, competency and outcome. If approved evidence is insufficient, it withholds the answer or directs the learner to suitable support.

EduScape also supports short formative activities and practice. Learner interactions contribute evidence that can reveal strengths and gaps, guide explanations and recommend the next appropriate activity. These are learning-support estimates, not official declarations of mastery; lecturers and institutions retain final academic authority. All model inference will use open-source models on institution-controlled infrastructure. No public AI service will process protected course or learner content.

2.3 Evidence of Efficacy and Feasibility

EduScape is an advanced working prototype, not an untested idea. It already demonstrates student, lecturer and administrator workflows; course-offering access control; course-grounded AI; structured course specifications and roadmaps; assessment, grading and progress functions; and HA-RAG retrieval. Clean-environment verification has also established a reproducible technical and security baseline.^[6]

In early sessions with 12 FCI students and four lecturers, all 12 students reported that source-anchored answers increased trust; three lecturers said they would approve the system if lecturers retained exclusive control over ingestion; and two MUST ICT staff expressed conditional support subject to an integration audit.^[7] These are promising feasibility and desirability signals, not proof of institution-wide impact. The proposed controlled pilot will produce stronger quality, security, usability, workload and operating-cost evidence.

2.4 Competitive Landscape Analysis Table (CLAT)

Feature	Moodle Core AI	EduGears AI	EduTelligence / Atlas	AI Learn	Public AI	EduScape AI
LMS integration	✓	✓	✓	✓	×	✓
Course-grounded answers	×	✓	✓	✓	×	✓
Source attribution	×	×	×	✓	×	✓
CBE course roadmap	×	×	×	×	×	✓
Competency/resource mapping	×	×	×	×	×	✓
Roadmap-aware AI	×	×	×	×	×	✓
Hierarchical retrieval	×	×	×	×	×	✓
Course-level access control	×	×	×	×	×	✓
Competency recommendations	×	×	✓	×	×	✓
Evidence-based refusal	×	×	×	✓	×	✓
Lecturer-controlled assessment	✓	✓	×	✓	×	✓
Multi-LMS architecture	×	×	×	×	×	✓

3. Commercial Viability

3.1 Business Model Canvas

KEY PARTNER-SHIPS • CITT and the Software Incubation Unit • MUST ICT Services for LMS access, infrastructure and technical integration • Participating faculties, lecturers and students for product validation • MUST Quality Assurance and academic leadership • Future LMS, deployment and replication partners	KEY ACTIVITIES • Developing, testing and maintaining EduScape • Integrating EduScape with institutional LMS platforms • Digitising CBE frameworks and configuring course roadmaps • Providing secure, source-grounded and roadmap-aware AI support • Managing governance, training, monitoring, sales and renewals	VALUE PROPOSITION • Helps institutions operationalise CBE without replacing their LMS • Gives students timely guidance supported by lecturer-approved materials • Links guidance to relevant competencies and learning outcomes • Helps lecturers create editable CBE course roadmaps and reduces repeated support work • Keeps content, learner data, AI inference and academic decisions under institutional control	CUSTOMER RELATIONSHIPS • Consultative institutional onboarding and LMS integration • Lecturer, administrator and ICT staff training • Dedicated institutional account management and helpdesk support • Regular reports, annual reviews, feedback and continuous improvement	CUSTOMER SEGMENTS • Initial customers: Ugandan universities using an LMS and implementing CBE • Buyers: University management, faculties and departments • Decision-makers: ICT, Quality Assurance, deans and academic leaders • Users: Lecturers and students • Future customers: Other tertiary institutions, schools, training providers and international institutions
	KEY RESOURCES • Technical, academic and commercial team • EduScape software, CBE roadmap model and HA-RAG know-how • Institution-controlled computing and open-source AI models • Secure infrastructure, LMS access, approved materials and evaluation data		CHANNELS • Direct institutional presentations and demonstrations • Pilots through MUST, CITT and participating faculties • University procurement and partnership agreements • Conferences, referrals and future LMS partners	
COST STRUCTURE Fixed costs: Product development, technical staff, platform maintenance, security, compliance and core infrastructure. Variable costs: AI inference, electricity, hosting, storage, backups, LMS integration, onboarding, training, support, monitoring, evaluation and sales.			REVENUE STREAMS Annual active-seat subscriptions; one-time LMS integration and onboarding fees; CBE and roadmap configuration; managed private deployments; training services; additional storage, support and AI-usage packages; and future partner licensing.	

3.2 Cost of Production

The UGX 21 million is the one-time cost of converting the prototype into a pilot-ready desktop/web product and validating it. Major GPU hardware and a dedicated mobile application are excluded. Open-source inference will run on available institution-controlled infrastructure; formal infrastructure access is a Month 1 delivery gate.

Period and output	Amount (UGX)
Month 1: audit, approvals, pilot specifications and Moodle/inference design	1,500,000
Months 2–3: pilot-ready CBE, learning-domain, HA-RAG, web-interface and Moodle development	9,400,000
Month 4: integration/security testing, training, benchmarking and release fixes	4,100,000
Month 5: controlled hosting, local inference, support, monitoring and pilot operations	3,500,000
Month 6: evaluation, handover, buyer interviews, IP/business review and PQR demo	2,500,000
Total	21,000,000

For a 100-student/six-lecturer deployment, preliminary annual recurring cost is approximately UGX 2.4 million: UGX 1.05 million for institution-controlled model operation, electricity and contingency; UGX 0.6 million for shared hosting, storage and backups; UGX 0.25 million for monitoring and security operations; and UGX 0.5 million for support. Customer-specific onboarding is estimated at UGX 3.5 million. These assumptions will be replaced with measured pilot costs.

3.3 Pricing Strategy

EduScape will use annual subscriptions paid in advance by the institution, a familiar education-software model.^[12] Pricing is based on active seats and includes defined allowances for course usage, storage, support and AI usage. A separate one-time onboarding fee covers LMS configuration, CBE and course mapping, training, security checks and acceptance testing.

Three tiers are proposed. The **Course Pilot** tier covers up to 250 students and 20 lecturers and is priced at US\$15 per student and US\$20 per lecturer per year, with one-time onboarding of UGX 5 million. The **Faculty** tier covers 251 to 1,500 students and is priced at US\$12 per student and US\$18 per lecturer per year, with one-time onboarding of UGX 7.5 million. The **Institution** tier applies above 1,500 students and uses a negotiated volume rate with a minimum annual commitment, with one-time onboarding of UGX 15 million. These tiers keep entry costs low for small pilots while allowing larger institutions to benefit from volume pricing.

For the proposed MUST pilot, 100 student seats and six lecturer seats produce an annual subscription of about UGX 5.87 million at the planning rate of UGX 3,625 per US\$. Adding the standard UGX 5 million onboarding fee gives a first-year value of about UGX 10.87 million. The university is the payer, and any internal cost-recovery arrangement is its own decision. Campus-wide pricing will use volume discounts rather than simply multiplying the pilot rate across every student.

3.4 Market Sizing

Market	Basis	Illustrative annual value
TAM	146,484 registered Moodle sites worldwide ^[10] × the minimum pilot subscription; excludes Canvas, Open edX and other LMS sites	UGX 860.23 billion
SAM	75 licensed Ugandan universities × the minimum pilot subscription	UGX 440.44 million
SOM	MUST plus two further universities within three years	UGX 17.62 million recurring, plus about UGX 15 million onboarding

4. Team and Partnerships

4.1 Core Team

Team member	Role and contribution
Twesigye John Davidson — MUST FCI student	Team Lead; AI pipeline, HA-RAG architecture, backend integration, evaluation and technical reporting
Kagyeza Eduardo — MUST FCI student	Systems and Integration Lead; web interface, deployment, Moodle mapping, monitoring and support
Ayebare Patience Praise — MUST FCI student	AI Interaction Specialist; benchmark design, learner interaction and user support
Ninsiima Doreen — MUST alumna	Business Development and Commercialisation Lead; customer discovery, pricing and market development
Rumwaya Gabriel Irumba — MUST alumnus	Industry Adviser; product strategy and private-sector partnerships
Dr Richard Kimera — MUST FCI staff	Academic Supervisor; research design, academic-quality review, institutional coordination and interpretation of results

4.2 Strategic Partnerships

CITT provide the incubation home and proposed grant, with support for planning, accountability, IP and commercialisation. MUST ICT Services is the proposed technical integration partner and has verbally agreed to support access to an isolated Moodle test environment; a formal agreement, permissions and infrastructure responsibilities remain Month 1 requirements. The participating faculties, six lecturers, two offerings and 100 students are proposed pilot participants and will be confirmed after award and approvals. MUST Quality Assurance is a prospective reviewer of CBE and academic quality. NCHE, NCDC and the Ministry of Education and Sports are stakeholders rather than confirmed implementation partners. Bishop Stuart and Kabale Universities are proposed later replication and buyer-validation sites, subject to separate engagement.

References

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Pitch deck Video: [EduScapeAI Pitch Deck](#).